

Dancing on the Saddles: A Geometric Framework for Stochastic Equilibrium Dynamics¹

Supplemental Material

A Proof of Proposition 4 (aggregate uncertainty and the conditional steady states)

Let $R(A, K) := 1 - \delta + \alpha AK^{\alpha-1}$ denote the gross return, $c(K, A)$ the stochastic-equilibrium consumption function, and $c^{nul}(A, K) := AK^\alpha - \delta K$ the conditional K -nullcline (shorthand for the main text's $C_A^{Kcnull}(K)$). As in the conditional steady-state Euler equation of Section 2.2, define the conditional Euler residuals at candidate K -stationary points, using equilibrium post-switch consumption $c(K, A')$:

$$\begin{aligned} F_B(K) &:= \beta \left[\pi_{BB} R(B, K) + \pi_{BG} \frac{c^{nul}(B, K)}{c(K, G)} R(G, K) \right] - 1, \\ F_G(K) &:= \beta \left[\pi_{GG} R(G, K) + \pi_{GB} \frac{c^{nul}(G, K)}{c(K, B)} R(B, K) \right] - 1. \end{aligned}$$

By that Euler equation, $F_B(K_B^{cs}) = 0$ and $F_G(K_G^{cs}) = 0$.

Proposition A.1 (Aggregate uncertainty and the conditional steady states; restatement of Proposition 4).

Suppose that (i) at each regime's PF steady state, the consumption response to a regime switch dominates the return response (jump dominance),

$$\frac{c(K_B^{pf}, G)}{c_B^{pf}} > \frac{R(G, K_B^{pf})}{R(B, K_B^{pf})}, \quad \frac{c(K_G^{pf}, B)}{c_G^{pf}} < \frac{R(B, K_G^{pf})}{R(G, K_G^{pf})};$$

and (ii) each conditional Euler residual F_A defined above is strictly positive for $K < K_A^{cs}$ and strictly negative for $K > K_A^{cs}$ on the relevant range (single crossing). Then

$$K_B^{cs} < K_B^{pf} < K_G^{pf} < K_G^{cs}, \quad c_B^{cs} < c_B^{pf} < c_G^{pf} < c_G^{cs}.$$

Proof.

At any conditional or PF steady state, consumption lies on the nullcline: $c_A^{cs} = c(K_A^{cs}, A) = c^{nul}(A, K_A^{cs})$ and $c_A^{pf} = c^{nul}(A, K_A^{pf})$.

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The PF steady states solve the Euler equations under absorbing beliefs, $1 = \beta R(B, K_B^{pf})$ and $1 = \beta R(G, K_G^{pf})$. Since $R(A, K)$ is strictly increasing in A and strictly decreasing in K (as $\alpha - 1 < 0$), immediately $K_B^{pf} < K_G^{pf}$; this step is belief-free and uses neither hypothesis. Hypothesis (ii) makes K_B^{cs}, K_G^{cs} the unique zero crossings of F_B, F_G , from above.

Step 1: show $K_B^{cs} < K_B^{pf}$. Evaluate F_B at K_B^{pf} , using $1 = \beta R(B, K_B^{pf})$, $\pi_{BB} = 1 - \pi_{BG}$, and $c^{nul}(B, K_B^{pf}) = c_B^{pf}$:

$$\begin{aligned} F_B(K_B^{pf}) &= \beta \left[(1 - \pi_{BG}) R(B, K_B^{pf}) + \pi_{BG} \frac{c_B^{pf}}{c(K_B^{pf}, G)} R(G, K_B^{pf}) \right] - 1 \\ &= \pi_{BG} \left[\beta \frac{c_B^{pf}}{c(K_B^{pf}, G)} R(G, K_B^{pf}) - 1 \right]. \end{aligned}$$

Jump dominance gives $c(K_B^{pf}, G) > c_B^{pf} R(G, K_B^{pf}) / R(B, K_B^{pf})$, hence

$$\beta \frac{c_B^{pf}}{c(K_B^{pf}, G)} R(G, K_B^{pf}) < \beta R(B, K_B^{pf}) = 1,$$

so the bracket is strictly negative and $F_B(K_B^{pf}) < 0$. As F_B is negative only above its zero, $K_B^{cs} < K_B^{pf}$.

Step 2: show $K_G^{cs} > K_G^{pf}$. Symmetrically, using $1 = \beta R(G, K_G^{pf})$, $\pi_{GG} = 1 - \pi_{GB}$, and $c^{nul}(G, K_G^{pf}) = c_G^{pf}$,

$$F_G(K_G^{pf}) = \pi_{GB} \left[\beta \frac{c_G^{pf}}{c(K_G^{pf}, B)} R(B, K_G^{pf}) - 1 \right].$$

Jump dominance gives $c(K_G^{pf}, B) < c_G^{pf} R(B, K_G^{pf}) / R(G, K_G^{pf})$, hence

$$\beta \frac{c_G^{pf}}{c(K_G^{pf}, B)} R(B, K_G^{pf}) > \beta R(G, K_G^{pf}) = 1,$$

so $F_G(K_G^{pf}) > 0$. As F_G is positive only below its zero, $K_G^{pf} < K_G^{cs}$.

Step 3: conclude the ordering and translate to consumption. Steps 1 and 2 give $K_B^{cs} < K_B^{pf}$ and $K_G^{pf} < K_G^{cs}$, and already $K_B^{pf} < K_G^{pf}$, hence $K_B^{cs} < K_B^{pf} < K_G^{pf} < K_G^{cs}$. The K -nullcline $c^{nul}(A, K) = AK^\alpha - \delta K$ is strictly increasing in A and increasing in K on the dynamically efficient region ($\alpha AK^{\alpha-1} > \delta$, which holds at all four steady

states under standard calibrations), so the same ordering carries over:

$$c_B^{cs} < c_B^{pf} < c_G^{pf} < c_G^{cs}.$$

■

Reading the hypotheses Jump dominance compares two responses to the same regime switch at a PF steady state: the return ratio deviates from one only at first order in the regime gap and with a small coefficient, $R(G, K)/R(B, K) = 1 + (G - B) \alpha K^{\alpha-1}/R(B, K)$, whereas equilibrium consumption responds to a persistent switch at first order with an $O(1)$ coefficient. Both hypotheses are checkable from the computed decision rules: (i) by evaluating $c(\cdot, A')$ at the two PF steady states, (ii) by evaluating F_A on a grid of K . In the baseline quarterly calibration ($\beta = 0.99$, $\alpha = 0.33$, $\delta = 0.025$, $A \in \{0.99, 1.01\}$, $\pi_{AA} = 0.90$) both hold with wide margins: at K_B^{pf} consumption rises by 0.77% upon a switch to G against a return-ratio threshold of 0.07%, and at K_G^{pf} it falls by 0.76%; each residual F_A crosses zero exactly once on the computed saddle range, from above. The implied orderings are confirmed in the computed equilibrium, $K_B^{cs} = 27.53 < K_B^{pf} = 27.93 < K_G^{pf} = 28.77 < K_G^{cs} = 29.22$ (the PF-comparison figure in Section 2.2).

Necessity of the strict conditions The hypotheses cannot be dropped. Under full depreciation ($\delta = 1$) with log utility and Cobb–Douglas production, the equilibrium policy takes the Brock–Mirman form $K' = \alpha \beta A K^\alpha$ with $c = (1 - \alpha \beta) A K^\alpha$, independently of Π . Upon a switch at fixed K , consumption then moves exactly by G/B , and so does $R(A, K) = \alpha A K^{\alpha-1}$: jump dominance holds with equality, and $K_A^{cs} = K_A^{pf} = (\alpha \beta A)^{1/(1-\alpha)}$. The strict orderings therefore genuinely require strict jump dominance; away from this degenerate corner the conditions hold in the calibrated equilibrium and the strict inequalities obtain.

B Proof of Proposition 6 (state-dependent peak timing)

Proof.

We derive the peak time in closed form and differentiate it to sign the comparative static in initial capital.

Step 1 (Blanchard–Kahn solution on the saddle branch). Log-linearize the RBC dynamics around the non-stochastic steady state, with $\hat{k}_t := \ln K_t - \ln K^*$ and $\hat{a}_t := \ln A_t - \ln A^*$. The linearized capital law of motion and saddle-path consumption policy are

$$\hat{k}_{t+1} = \lambda \hat{k}_t + \theta \hat{a}_t, \quad (1)$$

$$\hat{c}_t = \psi_k \hat{k}_t + \psi_a \hat{a}_t, \quad (2)$$

where $\lambda \in (0, 1)$ is the stable eigenvalue, $\theta > 0$ the capital-TFP elasticity, and $\psi_k > 0$, $\psi_a > 0$ the consumption elasticities on the saddle branch. For a one-time shock $\hat{a}_0 = \varepsilon > 0$ with $\hat{a}_t = \rho^t \varepsilon$, $\rho \in (0, 1)$, iterating (1) gives (using $\lambda \neq \rho$)

$$\hat{k}_t = \lambda^t \hat{k}_0 + \theta \varepsilon \sum_{s=0}^{t-1} \lambda^{t-1-s} \rho^s = \lambda^t \hat{k}_0 + \frac{\theta \varepsilon}{\lambda - \rho} (\lambda^t - \rho^t). \quad (3)$$

Substituting into (2) yields $\hat{c}_t = \mathcal{P} \lambda^t + \mathcal{Q} \rho^t$, where

$$\mathcal{P}(\hat{k}_0) := \psi_k \left[\hat{k}_0 + \frac{\theta \varepsilon}{\lambda - \rho} \right], \quad \mathcal{Q} := \left[\psi_a - \frac{\psi_k \theta}{\lambda - \rho} \right] \varepsilon. \quad (4)$$

Crucially, $\mathcal{P}$ depends on $\hat{k}_0$ while $\mathcal{Q}$ does not.

Step 2 (Peak time in closed form). Treating t as continuous, the peak satisfies $d\hat{c}_t/dt = 0$, i.e. $\mathcal{P} \ln \lambda \cdot \lambda^{t^*} + \mathcal{Q} \ln \rho \cdot \rho^{t^*} = 0$, so $(\lambda/\rho)^{t^*} = -\mathcal{Q} \ln \rho / (\mathcal{P} \ln \lambda)$. Taking logarithms,

$$t^* = \frac{1}{\ln(\lambda/\rho)} \ln \left(\frac{-\mathcal{Q} \ln \rho}{\mathcal{P} \ln \lambda} \right), \quad (5)$$

well-defined when $-\mathcal{Q} \ln \rho / (\mathcal{P} \ln \lambda) > 0$. Since $\ln \lambda < 0$ and $\ln \rho < 0$ cancel, positivity requires $\mathcal{Q}/\mathcal{P} < 0$; with $\mathcal{P} > 0$ this means $\mathcal{Q} < 0$, i.e. $\psi_a < \psi_k \theta / (\lambda - \rho)$: the direct TFP effect is dominated by the capital-accumulation channel, which is exactly what generates the hump.

Step 3 (Comparative static by direct differentiation). Differentiating (5), and using $\partial \mathcal{P} / \partial \hat{k}_0 = \psi_k > 0$ from (4),

$$\frac{\partial t^*}{\partial \hat{k}_0} = \frac{1}{\ln(\lambda/\rho)} \cdot \frac{-1}{\mathcal{P}} \cdot \frac{\partial \mathcal{P}}{\partial \hat{k}_0} = \frac{-\psi_k}{\mathcal{P}(\hat{k}_0) \cdot \ln(\lambda/\rho)}. \quad (6)$$

Under $\lambda > \rho$ we have $\ln(\lambda/\rho) > 0$; combined with $\psi_k > 0$ and $\mathcal{P} > 0$ (which holds when $\hat{k}_0 > -\theta\varepsilon/(\lambda - \rho)$, easily satisfied within the ergodic capital range), this gives $\partial t^*/\partial \hat{k}_0 < 0$. That is, the consumption peak arrives strictly later when initial capital is farther below the steady state. $\blacksquare$

C Proofs of Theorem 2 and Proposition 9 (synchronization and the across-spell bound)

Proof of Theorem 2 (synchronization under frozen-regime contraction)

Proof.

Part (i): existence of the long-history limit. Fix a regime history $\omega = (\dots, A_{t-2}, A_{t-1})$ and, for $T \geq 0$, let $G_T := \Gamma_{A_{t-1}} \circ \dots \circ \Gamma_{A_{t-T-1}} : \mathcal{U} \rightarrow \mathcal{U}$ (forward invariance). For $T' > T$ and $\Phi, \Phi' \in \mathcal{U}$, writing $\Psi := (\Gamma_{A_{t-T-2}} \circ \dots \circ \Gamma_{A_{t-T'-1}})(\Phi') \in \mathcal{U}$,

$$d(G_T(\Phi), G_{T'}(\Phi')) = d(G_T(\Phi), G_T(\Psi)) \leq \lambda_c^{T+1} \text{diam}(\mathcal{U}), \quad (7)$$

since G_T composes $T + 1$ maps each contracting at rate λ_c . Thus $\{G_T(\Phi)\}$ is Cauchy uniformly in Φ , and $\Phi^*(\omega) := \lim_{T \rightarrow \infty} G_T(\Phi)$ exists, is independent of Φ , and lies in $\mathcal{U}$ ($\mathcal{U}$ closed in the complete $(\mathcal{X}, d)$). Uniqueness: any $\tilde{\Phi} \in \mathcal{U}$ consistent with ω (i.e. $\tilde{\Phi} = G_T(\tilde{\Phi}_{t-T-1})$ for all T) satisfies $d(\tilde{\Phi}, \Phi^*(\omega)) \leq \lambda_c^{T+1} \text{diam}(\mathcal{U})$ for every T , hence $\tilde{\Phi} = \Phi^*(\omega)$; this is the random-fixed-point construction of [Schenk-Hoppé and Schmalfuß \(2001\)](#) for finitely many contracting maps selected by a Markov chain. With $\Gamma^{(T)} := \Gamma_{A_{t-1}} \circ \dots \circ \Gamma_{A_{t-T}}$, the identity $\Phi^*(\omega) = \Gamma^{(T)}(\Phi^*(\sigma^{-T}\omega))$ and rate- λ_c^T contraction give

$$d(\Gamma^{(T)}(\Phi), \Phi^*(\omega)) \leq \lambda_c^T \text{diam}(\mathcal{U}). \quad (8)$$

Part (ii): suffix synchronization. If ω, ω' agree over the most recent T periods with common composition $\Gamma^{(T)}$, then $\Phi^*(\omega) = \Gamma^{(T)}(\Phi^*(\sigma^{-T}\omega))$ (and likewise ω'), so with $\Phi^*(\sigma^{-T}\omega), \Phi^*(\sigma^{-T}\omega') \in \mathcal{U}$,

$$d(\Phi^*(\omega), \Phi^*(\omega')) \leq \lambda_c^T \text{diam}(\mathcal{U}). \quad (9)$$

$\blacksquare$

Proof of Proposition 9 (across-spell approximation bound)

Proof.

Step 1 (Pathwise bound). By Theorem 2(i), $\Phi = \Phi^*(\omega)$, $\Phi' = \Phi^*(\omega')$; if the histories agree over the most recent T periods, Theorem 2(ii) gives $d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U})$. This pathwise bound survives conditioning on any subset of realizations, in particular fixing a common K , so it holds for every pair in the same- K cloud.

Step 2 (Distribution of the common suffix length). Draw the two histories independently from the stationary regime chain, conditioned to agree at the current date ($T \geq 1$). The symmetric two-state chain has uniform stationary distribution, hence is reversible with the same reversed transition matrix. Stepping backward, the two reversed chains still agree if both persist (ρ_A^2) or both switch ($(1 - \rho_A)^2$), independently across periods, so with

$$q := \rho_A^2 + (1 - \rho_A)^2 \in (1/2, 1), \quad (10)$$

$$\Pr(T = k) = q^{k-1}(1 - q), \quad k = 1, 2, \dots$$

Step 3 (Expected cloud diameter). Combining Steps 1–2,

$$\mathbb{E}[d(\Phi, \Phi')] \leq \text{diam}(\mathcal{U}) \sum_{k=1}^{\infty} q^{k-1}(1 - q)\lambda_c^k = \frac{(1 - q)\lambda_c}{1 - q\lambda_c} \text{diam}(\mathcal{U}), \quad (11)$$

convergent since $q\lambda_c < 1$. The factor is below one and vanishes as $\rho_A \rightarrow 1$ ($q \rightarrow 1$) or $\lambda_c \rightarrow 0$. ■

D Proof of Proposition 10 (local contraction rate on the conditional saddle)

Regularity conditions Fix $A \in \{B, G\}$ and assume at the conditional steady state Φ_A^{cs} :

- (i) Near Φ_A^{cs} , $\mathcal{X}$ is an open subset of a Banach space with norm $\|\cdot\|$ and $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$ is C^1 ; write $D\Gamma_A := D\Gamma_{\text{endo}}(\Phi_A^{cs}, A)$.
- (ii) The tangent to the conditional saddle $\mathcal{M}(\Phi_0, A)$ at Φ_A^{cs} is a one-dimensional invariant direction of $D\Gamma_A$: the span of an eigenvector with simple real eigenvalue $\lambda_A^s \in (0, 1)$, separated from the rest of the spectrum by a spectral gap.

- (iii) In a neighborhood of Φ_A^{cs} there is a C^1 one-dimensional locally invariant curve $\mathcal{W}_A \subset \mathcal{X}$, tangent at Φ_A^{cs} to that direction, with $\mathcal{M}(\Phi_0, A) \cap \mathcal{V} \subset \mathcal{W}_A$ for some neighborhood $\mathcal{V}$ of Φ_A^{cs} .
- (iv) The metric d is asymptotically equivalent to $\|\cdot\|$ near Φ_A^{cs} : $\lim_{\Phi, \Phi' \rightarrow \Phi_A^{cs}, \Phi \neq \Phi'} d(\Phi, \Phi') / \|\Phi - \Phi'\| = 1$.

Full statement Under (i)–(iv), for Φ, Φ' restricted to $\mathcal{M}(\Phi_0, A)$,

$$\lim_{\substack{\Phi, \Phi' \rightarrow \Phi_A^{cs} \\ \Phi, \Phi' \in \mathcal{M}(\Phi_0, A), \Phi \neq \Phi'}} \frac{d(\Gamma_{\text{endo}}(\Phi, A), \Gamma_{\text{endo}}(\Phi', A))}{d(\Phi, \Phi')} = |\lambda_A^s|, \quad (12)$$

and for every $\varepsilon > 0$ there is a neighborhood $\mathcal{U}_A^\varepsilon$ of Φ_A^{cs} on which $\Gamma_{\text{endo}}(\cdot, A)$ contracts $\mathcal{M}(\Phi_0, A)$ at rate $|\lambda_A^s| + \varepsilon$.

Aggregate-capital projection If in addition $K : \mathcal{X} \rightarrow \mathbb{R}$ is C^1 near Φ_A^{cs} , the curve $\mathcal{W}_A$ is tangent to $\text{span } e_A^s$ with e_A^s a unit stable eigenvector, and K is nondegenerate along that direction, then $\mathcal{W}_A$ and the orbit tail it contains are locally parametrized by K near K_A^{cs} , so (12) gives

$$K_{t+1} - K_A^{cs} = \lambda_A^s (K_t - K_A^{cs}) + o(|K_t - K_A^{cs}|),$$

exact as $K_t \rightarrow K_A^{cs}$: the form quoted in the main text.

Proof. Work in the Banach identification of (i); by (ii) let e_A^s be a unit eigenvector with $D\Gamma_A e_A^s = \lambda_A^s e_A^s$. By (iii) take a C^1 parametrization $\phi : (-\eta, \eta) \rightarrow \mathcal{W}_A$ with $\phi(0) = \Phi_A^{cs}$, $\phi'(0) = e_A^s$ carrying the orbit tail; for $\Phi = \phi(s)$, $\Phi' = \phi(s')$, $s \neq s'$,

$$\Phi - \Phi' = (s - s') e_A^s + r_1(s, s'), \quad \|r_1(s, s')\| = o(|s - s'|). \quad (13)$$

By (i), continuity of $D\Gamma_{\text{endo}}$ near Φ_A^{cs} gives the uniform two-point expansion $\Gamma_{\text{endo}}(\Phi, A) - \Gamma_{\text{endo}}(\Phi', A) = D\Gamma_A(\Phi - \Phi') + r_2$ with $\|r_2\| = o(\|\Phi - \Phi'\|)$. Substituting (13) and $D\Gamma_A e_A^s = \lambda_A^s e_A^s$ yields $\Gamma_{\text{endo}}(\Phi, A) - \Gamma_{\text{endo}}(\Phi', A) = \lambda_A^s (s - s') e_A^s + o(|s - s'|)$; taking norms with $\|\Phi - \Phi'\| = |s - s'| + o(|s - s'|)$ from (13),

$$\frac{\|\Gamma_{\text{endo}}(\Phi, A) - \Gamma_{\text{endo}}(\Phi', A)\|}{\|\Phi - \Phi'\|} \longrightarrow |\lambda_A^s| \quad \text{as } s, s' \rightarrow 0, \ s \neq s'. \quad (14)$$

By (iv), $d(\Phi, \Phi')/\|\Phi - \Phi'\| \rightarrow 1$ applied to numerator and denominator of (14) gives (12). For the neighborhood statement, fix $\varepsilon > 0$; the limit yields $\delta_\varepsilon > 0$ such that $\Phi, \Phi' \in \mathcal{M}(\Phi_0, A)$ with $d(\Phi, \Phi_A^{cs}), d(\Phi', \Phi_A^{cs}) < \delta_\varepsilon$ imply $d(\Gamma_{\text{endo}}(\Phi, A), \Gamma_{\text{endo}}(\Phi', A)) \leq (|\lambda_A^s| + \varepsilon) d(\Phi, \Phi')$; take $\mathcal{U}_A^\varepsilon := B_{\delta_\varepsilon}(\Phi_A^{cs})$. $\blacksquare$

E Primitive conditions for Assumption 2

Independent route to Assumption 2 from static primitives at $\mu = 1$ (rational expectations), without the homotopy of Appendix F: side-switching across the K -nullcline is excluded under MPC bounds and weak price feedback. Near the conditional steady state local spectral regularity alone gives exact K -sufficiency (part (a)); away from it the log–Cobb–Douglas structure reduces the verification chain to explicit interval inequalities organized by a K -monotonicity interval and a K -admissible continuation, checked on a computed continuation rather than from primitives (part (b)).

Two objects organize the hypotheses. An interval $[\underline{K}, \overline{K}]$ with $0 < \underline{K}$ and $\overline{K} \leq (A/\delta)^{1/(1-\alpha)}$ is a K -monotonicity interval with margin $\kappa \in (0, 1)$ if $B_A(\underline{K}) - N_A(\overline{K}) \leq 1 - \kappa$ and $2f(\overline{K}; A) + (1 - \delta)\overline{K} < 8\kappa/M_A(\underline{K})$, where B_A, N_A, M_A are the MPC–price–feedback, nullcline-slope, and nullcline-curvature functions given in the proof. A frozen- A continuation is K -admissible on $(\mathcal{K}, \mathcal{V}_A)$ if (i) its K -path and the segments joining consecutive (K_t, K_{t+1}) stay in $\mathcal{K}$, and (ii) every K -stationary state with capital in $\mathcal{K}$ lies in $\mathcal{V}_A$.

Theorem 1 (Within-spell K -sufficiency under log utility and Cobb–Douglas).

Fix $A \in \{B, G\}$ and a conditional saddle $\mathcal{M}(\Phi_0, A)$ in the heterogeneous-household economy of this section.

- (a) Suppose the regularity conditions of Proposition 10 hold at Φ_A^{cs} with stable eigenvalue $\lambda_A^s \in (0, 1)$. Then there exists a neighborhood $\mathcal{V}_A$ of Φ_A^{cs} such that, along the frozen- A continuation started from any $\Phi'_0 \in \mathcal{M}(\Phi_0, A) \cap \mathcal{V}_A$: (i) $\{K_t\}$ is strictly monotone and converges to K_A^{cs} without crossing it; (ii) $\Delta K_t = 0$ only at $\Phi_t = \Phi_A^{cs}$; and (iii) K is an exact conditional sufficient statistic on $\mathcal{M}(\Phi_0, A) \cap \mathcal{V}_A$.
- (b) Suppose, in addition, that Assumptions 3 and 4 hold, and let $\mathcal{K}$ be a K -monotonicity interval with margin $\kappa \in (0, 1)$. Then along every continuation that is K -admissible on $(\mathcal{K}, \mathcal{V}_A)$, with $\mathcal{V}_A$ as in part (a), Assumption 2 holds,

$\{K_t\}$ is strictly monotone until convergence, and K is an exact conditional sufficient statistic on the entire continuation.

Part (a) (proximity measured in the distributional state space) covers the late phase of a spell, part (b) the full ergodic range; under the [Krusell and Smith \(1998\)](#) calibration the ergodic range is a K -monotonicity interval with margin $\kappa = 0.95$ (verified below). Part (b) is not assumption-free: Assumption 4's derivation from Assumption 3 is open, and admissibility (ii) (no plateau outside the local neighborhood) is the single residual condition separating it from an unconditional global theorem.

Household MPC bounds Let $c(a, z; r, w) = \tilde{c}(x, z)$ be the stationary consumption policy, entering prices only through cash-on-hand $x = wz + (1 + r)a$. Along a frozen-regime continuation the relevant bound is Lipschitz in price-induced cash-on-hand changes, not a transfer MPC.

Assumption 3 (Bounded price-income MPC).

Along the frozen-regime continuation under A , there exists $\bar{m} \in (0, 1]$ such that, for all (a, z) in the support of Φ_t , all t , and all price-induced cash-on-hand levels $x' \geq x$ along the continuation,²

$$0 \leq \tilde{c}(x', z) - \tilde{c}(x, z) \leq \bar{m}(x' - x). \quad (15)$$

Monotonicity of consumption and saving in x ([Carroll and Kimball, 1996](#)) gives $0 \leq \tilde{c}(x', z) - \tilde{c}(x, z) \leq x' - x$, so the conservative $\bar{m} = 1$ of the Level-1 check needs no calibration or smoothness.

Define $\bar{Z} := \sup_{t \geq 0} \int z d\Phi_t$, $\bar{a} := \sup_{t \geq 0} \int a d\Phi_t$, and the *segment hull* $\mathcal{K}^{\text{seg}} := \bigcup_{t \geq 0} [\min\{K_t, K_{t+1}\}, \max\{K_t, K_{t+1}\}]$.

Two-step verification of Assumption 2 Lemma [E.1](#) gives a sufficient condition for *endpoint no-side-switching*, $H_t H_{t+1} > 0$ for every transition with $K_{t+1} \neq K_t$; Lemma [E.2](#) upgrades this to the full line-segment no-crossing of Assumption 2 via a curvature-margin check. Both are numerically checkable.

²Bounds *price-induced* income changes, not the transfer MPC; the Lipschitz form accommodates kinks from discrete idiosyncratic states, and precautionary saving pushes $\bar{m}$ below one ([Carroll and Kimball, 1996](#)). [Kaplan and Violante \(2014\)](#) estimate quarterly MPCs of 0.15–0.25, though nothing here depends on these values.

In the Krusell–Smith Cobb–Douglas benchmark $w_A(K) = (1 - \alpha)AK^\alpha$, $r_A(K) = \alpha AK^{\alpha-1} - \delta$, so $w'_A(K) = (1 - \alpha)\alpha AK^{\alpha-1}$, $r'_A(K) = \alpha(\alpha - 1)AK^{\alpha-2}$, $(C_A^{Knull})'(K) = \alpha AK^{\alpha-1} - \delta$. The first step needs a path-level finite-difference bound.

Assumption 4 (Pathwise finite-difference bound on aggregate consumption).

Along the frozen- A continuation, there exists $\bar{m} \in (0, 1]$ such that for every t with $K_{t+1} \neq K_t$,

$$|C_{t+1} - C_t| \leq \bar{m}(w'_A(\xi_t)\bar{Z} + |r'_A(\xi_t)|\bar{a})|K_{t+1} - K_t| \quad (16)$$

for some ξ_t between K_t and K_{t+1} .

Assumption 4 is the discrete-time aggregate analogue of Assumption 3 but logically distinct: it additionally requires distributional-composition and continuation-value effects not to exceed the static bound. The implied $\bar{m} \approx 0.03$ on the simulated Krusell and Smith (1998) path sits below the $\bar{m} = 1$ envelope of Proposition E.1 (gap reflecting the small constrained mass); deriving it from Assumption 3 is left to future work.

Lemma E.1 (Primitive condition for endpoint no-side-switching).

Fix $A \in \{B, G\}$ and the frozen- A continuation with $K_t := K(\Phi_t)$, $C_t := C(\Phi_t, A)$. Suppose Assumption 4 holds and

$$\sup_{K \in \mathcal{K}^{\text{seg}}} \bar{m}(w'_A(K)\bar{Z} + |r'_A(K)|\bar{a}) - \inf_{K \in \mathcal{K}^{\text{seg}}} (C_A^{Knull})'(K) < 1. \quad (17)$$

Then endpoint no-side-switching holds: $H_t H_{t+1} > 0$ for every transition with $K_{t+1} \neq K_t$. Strict monotonicity additionally requires unique K -stationarity (main text), ruling out nonterminal flat steps with $K_{t+1} = K_t$ at $\Phi_t \neq \Phi^{cs}$.

Proof.

With $H_t := C_t - C_A^{Knull}(K_t)$, $K_{t+1} - K_t = -H_t$. The mean-value theorem gives $C_A^{Knull}(K_{t+1}) - C_A^{Knull}(K_t) = N_t(K_{t+1} - K_t)$, $N_t := (C_A^{Knull})'(\xi_t^N)$, and (16) gives $|C_{t+1} - C_t| \leq B_t |H_t|$, $B_t := \bar{m}(w'_A(\xi_t)\bar{Z} + |r'_A(\xi_t)|\bar{a})$. Then $H_{t+1} = H_t(1 + N_t) + (C_{t+1} - C_t)$, so $|H_{t+1} - H_t(1 + N_t)| \leq B_t |H_t|$. Since $\xi_t, \xi_t^N \in \mathcal{K}^{\text{seg}}$, (17) gives $B_t - N_t < 1$, i.e. $1 + N_t - B_t > 0$; hence $H_{t+1} \geq H_t(1 + N_t - B_t) > 0$ for $H_t > 0$ and $H_{t+1} \leq H_t(1 + N_t - B_t) < 0$ for $H_t < 0$, so $H_t H_{t+1} > 0$ whenever $K_{t+1} \neq K_t$. Strict monotonicity further needs $K_{t+1} = K_t$ only at $\Phi_t = \Phi^{cs}(\Phi_0, A)$, by Assumption 1(i) (representative-agent) or the stable eigenvalue $\lambda_A^s \in (0, 1)$ (heterogeneous-agent). ■

Consecutive endpoints stay on the same side, but the straight segment can still cross the curved nullcline; the next lemma closes this gap.

Lemma E.2 (From endpoint no-side-switching to Assumption 2).

Let $N : K \mapsto N(K)$ be C^2 on $[K_t, K_{t+1}]$ with $|N''(K)| \leq M$, and set $H_\tau := C_\tau - N(K_\tau)$ for $\tau \in \{t, t+1\}$. If $H_t H_{t+1} > 0$ and

$$\min\{|H_t|, |H_{t+1}|\} > \frac{M(K_{t+1} - K_t)^2}{8}, \quad (18)$$

then the straight line segment from (K_t, C_t) to (K_{t+1}, C_{t+1}) does not intersect the graph of N .

Proof.

With L the segment and $G := L - N$, $G(K_t) = H_t$, $G(K_{t+1}) = H_{t+1}$, $|G''| \leq M$. Let L_H linearly interpolate $(K_t, H_t), (K_{t+1}, H_{t+1})$; the quadratic interpolation-error bound gives $\sup_{[K_t, K_{t+1}]} |G - L_H| \leq M(K_{t+1} - K_t)^2/8$. Since H_t, H_{t+1} share a sign, L_H is monotone with $|L_H| \geq \min\{|H_t|, |H_{t+1}|\}$, so (18) gives $|G - L_H| < |L_H|$ pointwise; thus G keeps that sign and $L \neq N$ on $[K_t, K_{t+1}]$. $\blacksquare$

Lemmas E.1 and E.2 together verify Assumption 2, both numerically checkable. In the baseline Krusell and Smith (1998) calibration $|H_t| = |K_{t+1} - K_t| \sim 10^{-2}$ and $M \sim 10^{-3}$, so $M(K_{t+1} - K_t)^2/8 \approx 10^{-8} \ll |H_t|$.

A local check near the conditional steady state The global bound (17) weakens to a neighborhood condition near the conditional steady state.

Corollary E.1 (A local check near K_A^{cs}).

Under Assumption 4, suppose for some $\varepsilon > 0$ the continuation has $K_t \in (K_A^{cs} - \varepsilon, K_A^{cs} + \varepsilon)$ for all large t and

$$\sup_{|K - K_A^{cs}| \leq \varepsilon} \bar{m}(w'_A(K) \bar{Z} + |r'_A(K)| \bar{a}) - \inf_{|K - K_A^{cs}| \leq \varepsilon} (C_A^{Knull})'(K) < 1. \quad (19)$$

Then $H_t H_{t+1} > 0$ for all $t \geq T$ (some $T < \infty$) with $K_{t+1} \neq K_t$. Moreover: (i) if unique K -stationarity also holds, $\{K_t\}$ is eventually strictly monotone; (ii) if additionally the curvature-margin condition of Lemma E.2 holds, Assumption 2 holds along the tail, equivalently for the continuation re-initialized at Φ_T .

Proof. Restrict Lemma E.1 to $|K - K_A^{cs}| \leq \varepsilon$. $\blacksquare$

Hierarchy of primitive sufficient conditions Proposition E.1 reduces (17) to a substitution at the conditional steady state under log utility and Cobb–Douglas; a numerical extension confirms CRRA in the empirically relevant range. With the perfect-foresight diagnostic (Proposition G.1) these form three levels: explicit pointwise bound (Level 1), numerical CRRA extension (Level 2), perfect-foresight-monotonicity fallback (Level 3).

Proposition E.1 (Level 1: pointwise bound under log utility and Cobb–Douglas). *Consider a heterogeneous-household economy with log utility, Cobb–Douglas production (capital share $\alpha \in (0, 1)$, depreciation $\delta \in (0, 1)$), and bounded supports for (z, a) . Let $R_A^{cs} := 1 + \alpha A(K_A^{cs})^{\alpha-1} - \delta$ and $r_A^{cs} := R_A^{cs} - 1 = \alpha A(K_A^{cs})^{\alpha-1} - \delta$. Under Assumption 3 with $\bar{m} = 1$, the left-hand side of (17) at $K = K_A^{cs}$ is*

$$\text{LHS}(K_A^{cs}) = (1 - \alpha)(r_A^{cs} + \delta)\bar{Z} + (1 - \alpha)(r_A^{cs} + \delta)\bar{a}/K_A^{cs} - r_A^{cs}, \quad (20)$$

so (17) holds pointwise at K_A^{cs} whenever (20) is below 1, extending to a neighborhood by continuity; for $\bar{m} < 1$ it is smaller still.

Proof.

At K_A^{cs} , using $\alpha A(K_A^{cs})^{\alpha-1} = r_A^{cs} + \delta$: $w'_A(K_A^{cs}) = (1 - \alpha)(r_A^{cs} + \delta)$, $|r'_A(K_A^{cs})| = (1 - \alpha)(r_A^{cs} + \delta)/K_A^{cs}$, $(C_A^{Knull})'(K_A^{cs}) = r_A^{cs}$. Substituting into (17) with $\bar{m} = 1$ gives (20); continuity extends the inequality to a neighborhood. ■

In the representative-agent case the Euler condition pins $R_A^{cs} = 1/\beta$; heterogeneity places market-clearing R_A^{cs} below this. Under stochastic regime-switch expectations R_A^{cs} instead solves $1 = \beta \mathbb{E}_A[(C_A^{cs}/C(\Phi_A^{cs}, A'))^\sigma R_{A'}(K_A^{cs})]$ at Φ_A^{cs} (with $R_{A'}(K_A^{cs}) := 1 + \alpha A'(K_A^{cs})^{\alpha-1} - \delta$), generally $\neq 1/\beta$; formula (20) applies in either case.

Proposition E.1 is pointwise plus continuity, not global: $\sup_{K \in \mathcal{K}^{\text{seg}}} \text{LHS}(K) < 1$ still needs numerical verification since $\bar{Z}, \bar{a}$ depend on the continuation.

Level 2: a numerical extension to CRRA utility Under CRRA with $\sigma \neq 1$, $\bar{m}(\sigma)$ in Assumption 3 has no clean analytical form. Numerically extending Level 1 for the Krusell and Smith (1998) calibration gives $\text{LHS}(K_A^{cs}) < 0.2$ for all $\sigma \in [0.5, 5]$, spanning the empirically relevant range; an analytical bound on the admissible σ -range remains open.

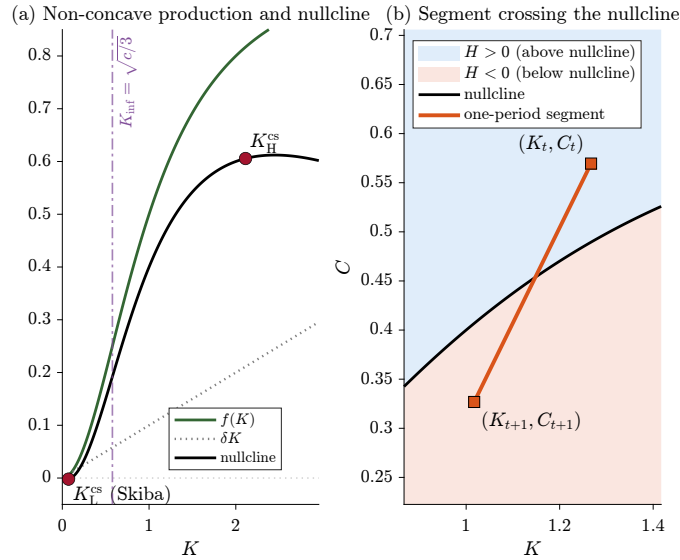
An illustration: non-concave production and the geometric possibility of segment crossing Assumption 2 is a genuine geometric restriction, not automatic from the budget constraint. For non-concave Hill-form production

$$f(K) = \frac{AK^2}{c + K^2}, \quad (21)$$

C^∞ with $f(0) = f'(0) = 0$ and inflection at $K = \sqrt{c/3}$, set $A = c = 1$, $\beta = 0.96$, $\delta = 0.10$; the Euler condition $f'(K) = 1/\beta - 1 + \delta$ has a low unstable root $K_L^{cs} \approx 0.07$ (convex region) and a Ramsey root $K_H^{cs} \approx 2.11$ (concave region). Figure E.1 shows the non-monotone nullcline, both steady states, and a budget-feasible segment ($K_{t+1} = f(K_t) + (1 - \delta)K_t - C_t$) that crosses it ($H_t > 0$, $H_{t+1} < 0$, so $H_t H_{t+1} < 0$).

This is a geometric possibility, not an equilibrium trajectory: whether the non-concave Ramsey policy selects such transitions requires solving under the non-convex value function. It shows only that non-concave technology *permits* crossings of the conditional K -nullcline that Assumption 2 excludes; in standard RBC and [Krusell and Smith \(1998\)](#) economies, concave Cobb–Douglas technology and bounded MPCs make (17) hold with slack.

Figure E.1: Non-concave production: the geometric possibility of segment crossing



Notes: Panel (a): $f(K) = AK^2/(c + K^2)$, $A = c = 1$, $\delta = 0.10$; dash-dotted line marks the inflection $K = \sqrt{c/3}$; steady states K_L^{cs} (unstable) and K_H^{cs} (Ramsey) on the nullcline $C = f(K) - \delta K$. Panel (b): a budget-feasible segment $(K_t, C_t) \rightarrow (K_{t+1}, C_{t+1})$ crossing the nullcline (blue: $H > 0$, $\Delta K < 0$; red: $H < 0$, $\Delta K > 0$), so $H_t H_{t+1} < 0$; only budget feasibility is illustrated.

Proof of Theorem 1 (within-spell K -sufficiency)

Proof.

Part (a): local result. Near the conditional steady state the distributional dynamics collapse to a one-dimensional linear recursion in the capital gap. Proposition 10 places the orbit tail in a C^1 invariant curve $\mathcal{W}_A$ tangent to the stable direction; with the aggregate-capital projection of Appendix D, (a) $\mathcal{W}_A$ is locally homeomorphically parametrized by K near Φ_A^{cs} , and (b) $k_t := K_t - K_A^{cs}$ obeys $k_{t+1} = \lambda_A^s k_t + o(|k_t|)$ with $\lambda_A^s \in (0, 1)$. Fix $\varepsilon := \frac{1}{2} \min\{\lambda_A^s, 1 - \lambda_A^s\} > 0$ and a neighborhood $\mathcal{V}_A$ of Φ_A^{cs} on which the parametrization applies and $|o(|k|)| \leq \varepsilon|k|$. For $\Phi'_0 \in \mathcal{M}(\Phi_0, A) \cap \mathcal{V}_A$ with $k_t \neq 0$, $(\lambda_A^s - \varepsilon)k_t \leq k_{t+1} \leq (\lambda_A^s + \varepsilon)k_t$ (reversed for $k_t < 0$); since $0 < \lambda_A^s - \varepsilon$ and $\lambda_A^s + \varepsilon < 1$, k_{t+1} shares the sign of k_t (so the path never crosses K_A^{cs} and stays in $\mathcal{V}_A$) and $|k_{t+1}| < |k_t|$, giving strict monotone geometric convergence to K_A^{cs} , proving (i), while $\Delta K_t = 0$ (i.e. $k_{t+1} = k_t$, so $\lambda_A^s k_t = k_t + o(|k_t|)$ with $\lambda_A^s \in (0, 1)$) forces $k_t = 0$, proving (ii). For (iii), the local homeomorphism makes $K_t \rightarrow K_A^{cs}$ imply $\Phi_t \rightarrow \Phi_A^{cs}$, and Theorem 1 with $e(\Phi) = K(\Phi)$ yields injectivity of K on the orbit piece, hence exact conditional sufficiency.

Part (b): explicit region under Assumptions 3–4. Under log utility and Cobb–Douglas the benchmark derivatives above identify the functions B_A, N_A, M_A : $B_A(K) = \bar{m}(w'_A(K)\bar{Z} + |r'_A(K)|\bar{a})$ (inside the sup of (17)), $N_A(K) = (C_A^{Knull})'(K)$ (inside the inf, so $B_A - N_A$ is the left side of (17)), $M_A(K) = |(C_A^{Knull})''(K)|$ (curvature in Lemma E.2). The cap $\bar{K} \leq (A/\delta)^{1/(1-\alpha)}$ ensures $C_A^{Knull} \geq 0$ on $[\underline{K}, \bar{K}]$; K -admissibility on $(\mathcal{K}, \mathcal{V}_A)$ keeps the K -path and all connecting segments in $[\underline{K}, \bar{K}]$ (condition (i)).

Step 1 (endpoint no-side-switching with margin). The mean-value points ξ_t (Assumption 4) and ξ_t^N (Lemma E.1) lie in $\mathcal{K}$; since B_A, N_A are decreasing, $B_t - N_t = B_A(\xi_t) - N_A(\xi_t^N) \leq B_A(\underline{K}) - N_A(\bar{K}) \leq 1 - \kappa$. Lemma E.1 then gives $H_t H_{t+1} > 0$ for $K_{t+1} \neq K_t$ and $|H_{t+1}| \geq (1 + N_t - B_t)|H_t| \geq \kappa|H_t|$.

Step 2 (segment no-crossing). Since M_A is decreasing, curvature is bounded by $M_A(\underline{K})$. Feasibility ($0 \leq C_t \leq f(K_t; A) + (1 - \delta)K_t$, $0 \leq C_A^{Knull} \leq f$) with the increasing $2f(K; A) + (1 - \delta)K$ and the second endpoint inequality gives $|H_t| = |K_{t+1} - K_t| \leq 2f(\bar{K}; A) + (1 - \delta)\bar{K} < 8\kappa/M_A(\underline{K})$. Then $\min\{|H_t|, |H_{t+1}|\} > M_A(\underline{K})|H_t|^2/8$: (a) $M_A(\underline{K})|H_t|/8 < \kappa \leq 1$ covers $|H_t|$; (b) $|H_{t+1}| \geq \kappa|H_t| > (M_A(\underline{K})|H_t|/8)|H_t|$. Steps 1–2 verify Assumption 2, and sign preservation makes $\{K_t\}$ strictly monotone.

Step 3 (sufficiency under admissibility (ii)). A strictly monotone bounded $\{K_t\}$,

or a flat step at finite t , gives a K -stationary limit with capital in $\mathcal{K}$; by (ii) it lies in $\mathcal{V}_A$ and by part (a)(ii) equals Φ_A^{cs} (terminal in the flat-step case). Either way $K_t \rightarrow K_A^{cs}$ strictly monotonically, and Theorem 1 with $e = K$ delivers exact conditional sufficiency. $\blacksquare$

Numerical verification under the Krusell and Smith (1998) calibration (Section 4.1 of the main text) With $\alpha = 0.36$, $\delta = 0.025$, $A \in \{0.99, 1.01\}$, $\bar{Z} \approx 1$, $\bar{a} \approx \sup_t K_t \approx 38.5$, $\bar{m} \leq 1$, $[K, \bar{K}] = [34.6, 38.5]$ (ergodic range): $B_A(\underline{K}) \approx 0.050$, $N_A(\bar{K}) \approx 0.010$, so $B_A(\underline{K}) - N_A(\bar{K}) \approx 0.040 < 0.05 = 1 - \kappa$ with $\kappa = 0.95$. For curvature, $M_A(\underline{K}) \approx 6.9 \times 10^{-4}$ gives $8\kappa/M_A(\underline{K}) \approx 1.1 \times 10^4$, while $2f(\bar{K}; A) + (1 - \delta)\bar{K} \approx 2(3.7) + 37.5 \approx 45$: a margin over two orders of magnitude. Here $\bar{Z} \leq z_{\max}$ is a priori from compact support; $\bar{a}$ and the ergodic range are read off the computed equilibrium, so the check validates the continuation rather than replacing computation. Replacing $\bar{a}$ by the feasibility cap $(A/\delta)^{1/(1-\alpha)} \approx 319$ still gives $B_A(\underline{K}) - N_A(\bar{K}) \approx 0.23 < 1$.

Why the residual condition resists primitive closure Admissibility condition (ii) requires no K -stationary state with capital in $\mathcal{K}$ to lie outside $\mathcal{V}_A$. Closing it from primitives means characterizing the basin of attraction of the conditional steady state, ruling out moving plateaus where aggregate net saving vanishes while Φ keeps evolving; the MPC apparatus controls aggregate consumption against capital but is silent when both vanish, so a primitive treatment would need either strict monotonicity of aggregate saving in a distributional ordering or a global Lyapunov function for $\Gamma_{\text{endo}}(\cdot, A)$, both beyond scope. The condition is checkable along any computed continuation and fails to bind in every calibration examined.

F Continuous dependence on beliefs

This appendix establishes two continuity results used by the PF diagnostic (Proposition G.1): continuity of the finite-horizon frozen-regime continuation $\{\Phi_t(\mu)\}_{t=0}^T$ from a given Φ_0 , which transports endpoint no-side-switching from $\mu = 0$ to $\mu = 1$; and continuity of the conditional steady state $\Phi_A^{cs}(\mu)$, which anchors its limit. Neither requires the ergodic distribution of the full stochastic process.

Proof of continuous dependence on μ .

Step 1 (Household policies). With $\Pi(\mu) = (1 - \mu)I + \mu\Pi$ entering the continuation ex-

pectation linearly, the Bellman operator T_μ for $v(a, z, A; \mu)$ on the compact $[\underline{a}, a_{\max}]$ is a sup-norm contraction for each μ under strict concavity and Inada (Stokey et al., 1989, Theorem 9.6) and is continuous in μ ; by the parametric contraction mapping theorem its fixed point $v(\cdot; \mu)$, and by the maximum theorem the policies $c(\cdot, A; \mu)$, $a'(\cdot, A; \mu)$, depend continuously on μ .

Step 2 (Frozen-regime operator). $\Gamma_{\text{endo}}(\cdot, A; \mu)$ composes the continuous price map $\Phi \mapsto (r(\Phi, A), w(\Phi, A))$, the jointly continuous Step 1 policy, and the push-forward to next-period distribution, so it is jointly continuous in (Φ, μ) (distributions under weak convergence, μ under Euclidean distance).

Step 3 (Finite iterates). For fixed Φ_0, T , $\Gamma_{\text{endo}}^t(\Phi_0, A; \mu)$ composes t Step 2 maps, so $\{\Phi_t(\mu)\}_{t=0}^T$ and the aggregates $\{(K_t(\mu), C_t(\mu))\}_{t=0}^T$ are continuous in μ pointwise in t for every finite T .

Step 4 (Conditional steady state). $\Phi_A^{cs}(\mu)$ solves $F(\Phi, \mu) := \Phi - \Gamma_{\text{endo}}(\Phi, A; \mu) = 0$. Assume at $\mu^* \in [0, 1]$: (i) $\Gamma_{\text{endo}}(\cdot, A; \mu^*)$ admits a well-defined linearization at $\Phi_A^{cs}(\mu^*)$; (ii) the stable eigenvalue $\lambda_A^s(\mu^*) \in (0, 1)$ is simple and real; (iii) 1 lies in the resolvent set of $D\Gamma_A(\mu^*)$ on the tangent space—for instance, $D\Gamma_A(\mu^*)$ is quasi-compact with essential spectral radius below one and 1 is not an eigenvalue—so that no part of the spectrum, point or essential, touches 1. Then $I - D\Gamma_A(\mu^*)$ is a bounded isomorphism on the tangent space, and the implicit function theorem (Step 2 supplies Fréchet differentiability of $\Gamma_{\text{endo}}(\cdot, A; \mu)$), with Step 2 joint continuity and continuity of the derivative in μ , gives a unique $\Phi_A^{cs}(\mu)$ near μ^* varying continuously in μ . ■

G Perfect-foresight diagnostic

K -monotonicity under rational expectations can be verified from the deterministic perfect-foresight model plus two regularity conditions (unique K -stationarity and uniform spectral regularity), because the K -nullcline is invariant over beliefs (Propositions 3 and 7).

Belief family For $\mu \in [0, 1]$, set $\Pi(\mu) := (1 - \mu)I + \mu\Pi$, with Π the true Markov matrix for $A \in \{B, G\}$: $\mu = 0$ is perfect foresight, $\mu = 1$ rational expectations. The μ -RCE is the recursive competitive equilibrium under $A' \sim \Pi(\mu)(\cdot|A)$. Fix (Φ_0, A) ; denote the frozen-regime continuation by $\{(K_t(\mu), C_t(\mu))\}_{t \geq 0}$ with conditional steady state $K_A^{cs}(\mu)$.

Since $\Pi(\mu)$ enters the Bellman equation linearly, the continuation and $K_A^{cs}(\mu)$ are continuous in μ by the parametric contraction mapping theorem and continuous dependence of the market-clearing fixed point (Appendix F). Define the *monotonicity set* $\mathcal{S} := \{\mu \in [0, 1] : \{K_t(\mu)\}_{t \geq 0} \text{ is strictly monotone along the frozen-}A \text{ continuation}\}$.

Definition G.1 (Regular belief homotopy).

Fix (Φ_0, A) . The belief homotopy is regular if, over $\mu \in [0, 1]$:

- (a) (Continuity) the family $\{(\Phi_t(\mu))_{t \geq 0}, \Phi_A^{cs}(\mu)\}$ is jointly continuous in μ ; and
- (b) (Uniform spectral regularity) at every μ , the linearized frozen-regime operator at $\Phi_A^{cs}(\mu)$ has a simple stable eigenvalue $\lambda_A^s(\mu) \in (0, 1)$, separated from the rest of the spectrum by a spectral gap bounded away from zero uniformly in μ .

Condition (a) is the continuity result of Appendix F; (b) imposes the spectral regularity of Proposition 10 (Appendix D) uniformly along the homotopy.

Proposition G.1 (Perfect-foresight diagnostic).

Fix (Φ_0, A) and suppose: (i) the perfect-foresight ($\mu = 0$) transition path from Φ_0 is strictly K -monotone; (ii) $\Delta K_t = 0$ only at $\Phi_t = \Phi_A^{cs}(\mu)$, for every $\mu \in [0, 1]$; and (iii) the belief homotopy is regular. Then aggregate capital is strictly K -monotone along the rational-expectations ($\mu = 1$) frozen-regime continuation.

Proof. $\mathcal{S}$ is nonempty ($0 \in \mathcal{S}$ by (i)), open, and closed in $[0, 1]$; by connectedness $\mathcal{S} = [0, 1]$. Openness uses nullcline invariance (Proposition 7) and continuity in μ on a finite window, with (iii) controlling the tail; closedness uses the same ingredients, with (ii) ruling out a limiting zero of the nullcline gap at a non-steady-state capital level. The openness step is detailed below. ■

Proposition G.1 reduces verification to a perfect-foresight monotonicity check plus the no-plateau and uniform-regularity conditions. The perfect-foresight check is textbook for the representative agent; for heterogeneous agents, monotone convergence of aggregate capital along computed Bewley–Aiyagari transitions is a robust numerical regularity (Lee, 2026). With the curvature-margin check of Appendix E, the proposition delivers the full segment condition of Assumption 2. It extends to any continuous aggregate e with unique stationarity, covering the CRRA–separable-labor extension.

Openness step (detail).

Tail monotonicity via eigenvalue continuity.

Fix $\mu^* \in \mathcal{S}$. Strict monotone convergence of $\{K_t(\mu^*)\}$ to $K_A^{cs}(\mu^*)$ forces the linearized frozen-regime map $K_{t+1} - K_A^{cs}(\mu^*) \approx \lambda_A^s(\mu^*)(K_t - K_A^{cs}(\mu^*))$ to have stable root $\lambda_A^s(\mu^*) \in (0, 1)$, since a negative or complex root would produce oscillatory local convergence. As the stable root of the linearized Euler–resource system at $K_A^{cs}(\mu)$, and since $K_A^{cs}(\mu)$ and the system coefficients depend continuously on μ (Appendix F) and eigenvalues depend continuously on entries, $\lambda_A^s(\mu)$ is continuous; hence there is $\varepsilon_1 > 0$ with $\lambda_A^s(\mu) \in (0, 1)$ for $|\mu - \mu^*| < \varepsilon_1$, giving monotone convergence near $K_A^{cs}(\mu)$. Choose T so that $K_t(\mu^*)$, $t \geq T$, lies in a neighborhood where the local map is order-preserving with slope in $(0, 1)$; by continuity of $(K_t(\mu), C_t(\mu))$ for each finite t , nearby $\{K_t(\mu)\}_{t \geq T}$ enters this neighborhood and inherits monotone convergence, while finite-horizon continuity preserves the sign of H_t for $t \leq T$. Thus monotonicity holds for all t and $\mu \in \mathcal{S}$. ■

H Calibrating the asymmetric adjustment cost

The asymmetric adjustment cost has two curvatures, μ_+ (expansion) and μ_- (contraction), reparameterized as a level $\bar{\mu} = (\mu_+ + \mu_-)/2$ and an asymmetry $r = \mu_+/\mu_-$; the other annual parameters are $\beta = 0.96$, $\delta = 0.10$, $\alpha = 0.33$, and $A \in \{0.98, 1.02\}$ with transition diagonal 0.90. Moments use annual log first-differences of real chained-dollar FRED series: private nonresidential fixed investment (PNFICA, 1967–2019), gross private domestic investment (GPDICA, 1947–2019), and GDP, all pre-COVID.

Asymmetry r . The friction makes contractions sharper than expansions, so the target is the negative skewness of investment growth. The third-moment coefficient is unreliable in a 52-observation sample dominated by 2009 (-1.03 full-sample, -0.54 dropping 2009, -0.32 for GPDI), so I target the outlier-robust downside/upside semideviation ratio σ_-/σ_+ (RMS below- over above-mean deviations of $\Delta \log I$): 1.32 (PNFI), 1.11 (GPDI), both above one. It rises monotonically with r in the model, so the mean-centered band implies $r \approx 2$ –3; I set $r = 3$, hence $(\mu_-, \mu_+) = (1.53, 4.57)$, with $\mathcal{R} \in [1.16, 1.34]$ across $r \in [2, 4]$ (1.27 at $r = 3$). This is a conditional discipline on the asymmetry, not a structural estimate.

Level $\bar{\mu}$. Relative investment volatility cannot identify the level: even at negligible friction the $\pm 2\%$ TFP process caps $\text{sd}(I)/\text{sd}(Y)$ near 2.75, below the data’s 3.0–4.5. I

anchor $\bar{\mu}$ on the investment- q elasticity: $q = 1 + \tilde{\mu}(K' - K)/K$ gives a net-investment-rate response to q of $1/\tilde{\mu}$, and at $\bar{\mu} = 3.05$, $r = 3$ the curvatures imply $1/\tilde{\mu} \in [0.22, 0.65]$, within measurement-corrected estimates. The level barely affects $\mathcal{R}$.

I Search and matching model

Monthly Diamond–Mortensen–Pissarides economy of Section 4.2: exogenous separation, Nash wages, Cobb–Douglas matching. A representative household holds a unit measure of workers; the employed produce z and earn w , the unemployed home-produce $b > 0$; firms post vacancies at cost $\kappa > 0$.

States and processes State $S = [u_{-1}, z]$ with predetermined $u_{-1} = 1 - n_{-1}$ and TFP

$$\ln z_t = \rho_z \ln z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_z^2), \quad (22)$$

discretized by Tauchen into 11 states at ± 3 s.d. Matching is

$$M(u, v) = m_0 u^\xi v^{1-\xi}, \quad (23)$$

with $u = 1 - n$ and tightness $\theta := v/u$, giving vacancy-filling

$$q(\theta) = \frac{M(u, v)}{v} = m_0 \theta^{-\xi}, \quad (24)$$

and job-finding $f(\theta) = m_0 \theta^{1-\xi}$ (q truncated at unity). Employment evolves as

$$n = (1 - \lambda) n_{-1} + q(\theta) v, \quad (25)$$

and consumption is output net of posting costs plus home production,

$$c(S) = n(S) z - \kappa v(S) + (1 - n(S)) b. \quad (26)$$

Preferences, wages, entry CRRA preferences give SDF $\Lambda(S, S') = \beta (c(S)/c(S'))^\sigma$. Nash bargaining with worker weight η yields

$$w(S) = (1 - \eta) b + \eta(z + \kappa \theta(S)), \quad (27)$$

and free entry

$$\frac{\kappa}{q(S)} = \beta \mathbb{E} \left[\left(\frac{c(S)}{c(S')} \right)^\sigma \left(z' - w(S') + (1 - \lambda) \frac{\kappa}{q(S')} \right) \middle| S \right]. \quad (28)$$

Timing is hybrid: current matches produce immediately in (25)–(26), but free entry values a vacancy only from next period, so its continuation $\kappa/q(S')$ carries survival weight $(1 - \lambda)$.

Recursive competitive equilibrium Policies $\{\theta, v, n, w, c\}(S)$ satisfying (28), (27), (25), (26) for all S with consistent expectations, solved globally via the repeated transition method (Lee, 2026).

Calibration Table I.1 lists the monthly calibration; notable choices are $\beta = 0.99^{1/3}$, the Hosios condition $\eta = \xi$, JOLTS $\lambda = 0.028$, and an 11-state Tauchen TFP chain at ± 3 s.d.

Table I.1: DMP model: calibrated parameters

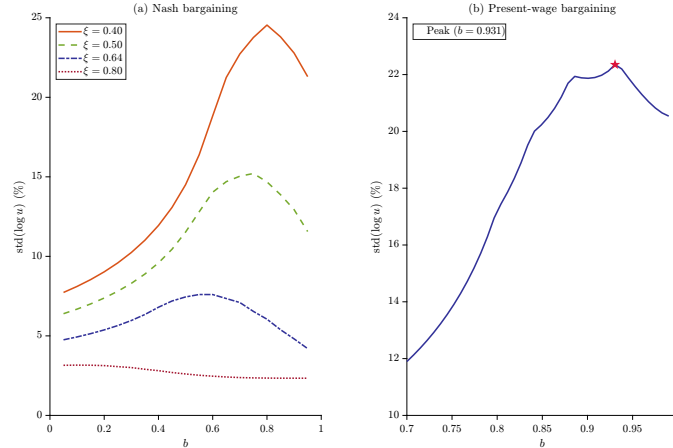
Parameter	Symbol	Value	Target / Source
Discount factor	β	$0.99^{1/3} \approx 0.9967$	Quarterly $\beta = 0.99$
Risk aversion	σ	1.000	Log utility
Matching efficiency	m_0	0.543	Steady-state job-finding rate
Matching elasticity	ξ	0.635	Petrongolo and Pissarides (2001)
Separation rate	λ	0.028	JOLTS monthly separation rate
Vacancy posting cost	κ	0.309	Steady-state vacancy rate
Home production	b	0.400	
Bargaining weight	η	0.635	Hosios condition ($\eta = \xi$)
TFP persistence	ρ_z	0.950	Monthly AR(1)
TFP innovation s.d.	σ_z	0.050	Output volatility
TFP grid points		11	Tauchen, ± 3 s.d.

Benefit extensions Two extensions of constant b . (i) Stochastic benefits $b \in \{b_L, b_H\}$ on a two-state chain ($\rho_b = 0.98$, $b_L = 0.400$, $b_H = b + 0.50 w^{ss}$) crossed with the TFP chain (22 states via Kronecker product); the four conditional saddles for $\{B, G\} \times \{b_L, b_H\}$ (B, G the ± 1 s.d. TFP states) order from (G, b_L) steepest to (B, b_H) flattest. (ii) Unemployment-dependent benefits $b(u) = b + \omega(u) \bar{b} w^{ss}$ with $\omega(u) = \min\{1, \max\{0, (u - u_\ell)/(u_h - u_\ell)\}\}$, thresholds $u_\ell = 0.055$, $u_h = 0.070$, scale

$\bar{b} = 1.50$: b rises from 0.400 at $u \leq u_\ell$ to $b + \bar{b}w^{ss}$ at $u \geq u_h$. Now two conditional saddles are indexed by TFP alone; the B regime settles at higher unemployment with compressed surplus while G keeps its slope, widening both the slope asymmetry and the steady-state unemployment gap.

Sensitivity of the hump-shaped volatility result Figure I.1 tests the non-monotone benefit-generosity/unemployment-volatility relation of Section 4.2. Panel (a) varies $\xi \in \{0.40, 0.50, 0.635, 0.80\}$ under Nash (Hosios $\eta = \xi$); panel (b) uses present-wage bargaining $w = (1 - \eta)b + \eta z$, removing the tightness-to-wage feedback. The hump persists for $\xi \leq 0.635$ (peak shifting left as ξ rises, nearly flat at $\xi = 0.80$) and under present-wage bargaining (peak at $b \approx 0.93$), so the non-monotonicity reflects surplus geometry, not a specific wage-feedback channel.

Figure I.1: Robustness of hump-shaped volatility: matching elasticity (panel a) and wage-setting (panel b)



Notes: Panel (a): $\text{std}(\log u)$ vs. b under Nash bargaining for $\xi \in \{0.40, 0.50, 0.635, 0.80\}$. Panel (b): $\text{std}(\log u)$ under present-wage bargaining, peaking at $b \approx 0.93$.

J Diagnostic validation in a lumpy-investment economy

This appendix runs the boundary test of Section 4.1, where nullcline invariance fails and K -sufficiency is no longer a theorem: does the monotonicity diagnostic still track scalar-state forecast accuracy? Three economies are solved with the same global method—Khan and Thomas (2008) in general equilibrium, the same model with prices frozen at stationary-equilibrium values (where lumpiness is known to matter), and

the [Krusell and Smith \(1998\)](#) benchmark. For each frozen-regime spell of length ≥ 3 I compute two statistics: the monotonicity diagnostic $|\sum_t \Delta K_t| / \sum_t |\Delta K_t|$, which is one iff capital moves monotonically within the spell (Proposition 8; spells with a ΔK sign change are *flagged*); and the scalar-state forecast error from a regime-specific rule $\log K' = b_0(A) + b_1(A) \log K$ iterated dynamically within the spell ([Den Haan, 2010](#)), recorded as the maximum log-deviation from the realized path.

Results The diagnostic validates both sides. On the success side, all 215 spells of the [Khan and Thomas \(2008\)](#) general-equilibrium economy and all 84 of the [Krusell and Smith \(1998\)](#) benchmark are exactly monotone in K (dynamic errors averaging 4×10^{-4} in logs), so price movements smooth aggregate investment enough to keep the within-spell path monotone—rationalizing approximate aggregation under lumpy investment. On the failure side, removing price feedback breaks monotonicity and accuracy together: interior monotonicity and flagged spells appear, mean dynamic errors rise thirtyfold (to 1.2×10^{-2}), and flagged spells carry errors about four times those of monotone spells, even as the one-step R^2 stays at 0.998 ([Den Haan, 2010](#)). Computable along any path at negligible cost, the diagnostic flags where scalar-state methods are solid and where the distribution carries independent forecasting power ([Bachmann et al., 2013](#)).

K Assumptions 1 and 2 in the representative-agent RBC

In the representative-agent RBC, Assumptions 1 and 2 follow from the scalar concave structure. The quantitative inputs are the Step 4 curvature-margin condition and, for stochastic beliefs, the Step 2c single-crossing condition on the conditional Euler residual. The endogenous state is scalar capital K , with frozen- A law of motion

$$g_A(K) := \Gamma_{\text{endo}}(K, A),$$

the equilibrium savings policy of the stochastic RCE with the regime held at A . Production is Cobb–Douglas $f(K; A) = AK^\alpha$, $\alpha \in (0, 1)$, depreciation $\delta \in (0, 1)$, utility strictly concave and Inada, and the regime a finite irreducible Markov chain.

Proposition K.1 (Geometric regularity holds in the canonical RBC).

Under the maintained Cobb–Douglas/strictly-concave-utility specification, together with

the curvature-margin condition of Step 4 and, for stochastic beliefs, the verified single-crossing condition of Step 2c, Assumptions 1 and 2 hold for the representative-agent RBC. For each A :

- (i) the frozen- A continuation converges to a unique conditional steady state K_A^{cs} , and aggregate capital is strictly K -monotone along it;
- (ii) the continuation makes no segment crossing of the conditional K -nullcline.

The curvature-margin condition is used only for (ii); the single-crossing condition only for the stochastic-beliefs case of (i). The deterministic-beliefs conclusions follow from strict concavity alone.

Proof.

Step 0 (Policy selection). The stochastic RCE solves a strictly concave dynamic program, so the optimal policy is single-valued and continuous (C^1 in the interior) in (K, A) (Stokey et al., 1989, Ch. 9). Restricting this policy to fixed A , g_A is a well-defined continuous increasing self-map of the capital domain, giving Assumption 1(ii).³

Step 1 (Feasibility rules out explosion). Any feasible continuation satisfies $K_{t+1} \leq f(K_t; A) + (1 - \delta)K_t$, and for $K_t > \bar{K}_A := (A/\delta)^{1/(1-\alpha)}$ diminishing returns give $f(K_t; A) < \delta K_t$, hence $K_{t+1} < K_t$. So $K_t \leq \max\{K_0, \bar{K}_A\}$ on every branch, a feasibility property needing no transversality condition.

Step 2 (Unique conditional steady state). (2a) Fixing $\underline{K} \in (0, \bar{K}_A)$ with $g_A(\underline{K}) \geq \underline{K}$ (exists by Inada), the increasing g_A maps $[\underline{K}, \bar{K}_A]$ into itself (Step 1), so a fixed point $K_A^{cs} = g_A(K_A^{cs})$ exists (Brouwer). (2b) As the policy of a strictly concave one-sector problem, g_A induces monotone convergence to a unique steady state: the turnpike property (Stokey et al., 1989, Ch. 6), i.e. $g_A(K) - K$ single-crosses zero. At $\mu = 0$ this is the deterministic Ramsey result. (2c) At $\mu = 1$ existence and monotonicity are inherited from concavity as in (2a)–(2b). Global uniqueness and monotone convergence need one further input: the conditional Euler residual at $\mu = 1$ crosses zero exactly

³Strict concavity, not transversality, rules out the frozen-branch overaccumulation path; transversality only selects the stable recursive policy. The necessary condition is the pooled (expectation) condition (Kamihigashi, 2003, 2005), refining to a regime-conditional one under a finite chain; neither disciplines the frozen- A branch, a probability-zero event. The all- A overaccumulation trajectory ($K_t \rightarrow \bar{K}$, $C_t \rightarrow 0$, $u'(C_t) \rightarrow \infty$) is capital-bounded and violates the branchwise condition yet is hidden in expectation by its weight π_{AA}^T ; strict concavity excludes it as suboptimal at every positive-probability finite-horizon node.

once on the computed range, so $g_A(K; 1) - K$ changes sign once. This single-crossing is not delivered by concavity; it is verified in the computed equilibrium (Appendix A: a single crossing from above of each conditional Euler residual at $\mu = 1$). The belief-continuity results of Appendix F give continuous variation of the steady state in μ , but uniqueness rests on the direct $\mu = 1$ check.

This gives Assumption 1(i), and unique K -stationarity follows because K is autonomous: $\Delta K_t = 0$ only at the fixed point.

Step 3 (Strict monotonicity; endpoint no-side-switching). With g_A increasing, slope $\lambda_A^s \in (0, 1)$ at K_A^{cs} , and Step 2 single-crossing, the iterates $\{K_t\}$ are monotone toward K_A^{cs} (from any K_0 under deterministic beliefs; from the verified range under stochastic beliefs). The identity $\Delta K_t = -H_t$, $H_t := C_t - C_A^{Knull}(K_t)$, then keeps H_t of constant nonzero sign before convergence: $H_t H_{t+1} > 0$.

Step 4 (Segment no-crossing; full Assumption 2). By the curvature-margin lemma (Lemma E.2), endpoint no-side-switching upgrades to segment no-crossing whenever $\min\{|H_t|, |H_{t+1}|\} > M(K_{t+1} - K_t)^2/8$, with $M = \sup \alpha(1 - \alpha)AK^{\alpha-2} < \infty$ on the compact saddle. The RBC identity $|K_{t+1} - K_t| = |H_t|$ reduces this to $|H_{t+1}| > M|H_t|^2/8$; near the steady state $|H_{t+1}| \approx \lambda_A^s |H_t|$ gives the linear bound $|H_t| < 8\lambda_A^s/M$. Since $|H_t| \rightarrow 0$, this holds along the tail, and globally the nullcline gap is feasibility-bounded (Step 1); it is an explicit curvature-margin condition satisfied by orders of magnitude under standard calibration. Steps 3–4 give Assumption 2. ■

The load-bearing input is strict concavity: it selects the policy (Step 0), bounds capital (Step 1), and yields the unique conditional steady state and monotone convergence (Steps 2–3); the stochastic case adds the verified single-crossing (Step 2c) and the feasibility-bounded curvature closing Step 4. Transversality enters only at selection and is silent on the frozen branch; feasibility, not transversality, rules out explosion. The heterogeneous-agent economy lacks this: Γ_{endo} acts on the infinite-dimensional distribution, K is no longer autonomous, and the scalar stable-eigenvalue property becomes a spectral property of an infinite-dimensional operator unreadable from primitives. The non-concave illustration of Appendix E confirms concavity is essential: dropping it produces budget-feasible transitions that cross the nullcline, so Assumption 2 can fail.

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